

Math 152 Problem Set 1

Due 10/7/2026, at 10:00 PM

Definition. Given a function f defined on any interval around a , but not necessarily at a itself, we say that $\lim_{x \rightarrow a} f(x) = L$ if for any $\varepsilon > 0$, there is a $\delta > 0$ such that if $0 < |x - a| < \delta$, then $|f(x) - L| < \varepsilon$.

Problem 1. In this problem, we will show that $\lim_{x \rightarrow 2} (-2x + 1) = -3$, using the definition of the limit.

- (a) Write down what the definition of the limit becomes when specialized to this particular case.
- (b) Rewrite the inequality $|-2x + 1 - (-3)| < \varepsilon$ as an inequality involving $|x - 2|$.
- (c) Based on part (b), given $\varepsilon > 0$, find a δ in terms of ε that works in the definition of the limit.

Problem 2. In this problem, we will verify that for any numbers a, b, c , we have $\lim_{x \rightarrow c} (ax + b) = ac + b$.

- (a) Write down what the definition of the limit becomes when specialized to this particular case.
- (b) Show that if $a \neq 0$, then given any $\varepsilon > 0$, taking $\delta = \varepsilon/|a|$ works.
- (c) Show that if $a = 0$, then for any $\varepsilon > 0$, any δ works.

Problem 3 (Optional). In this problem we will consider the form of the definition of the limit. Consider the following similar sounding statement:

There is a $\delta > 0$ such that for all $\varepsilon > 0$, if $0 < |x - a| < \delta$, then $|f(x) - L| < \varepsilon$.

Argue that this means that f is constant and equal to L on some interval around a (except maybe a itself). Thus this statement is much stronger than what the definition of the limit asserts.¹

Definition. Let f be a function defined on an interval around a . We say that f is continuous at a if $\lim_{x \rightarrow a} f(x) = f(a)$.

Problem 4. In this problem, we will show that the function $f(x) = x^3$ is continuous at 0.

- (a) Write down what it means for this function to be continuous at 0 in terms of a limit.
- (b) Write down what it means for this function to be continuous at 0 in terms of ε 's and δ 's.
- (c) Given an $\varepsilon > 0$, explain how to choose an appropriate δ to verify the statement in part (b).

Problem 5 (Optional). Argue from the ε, δ definition of continuity that if f is continuous at a and g is continuous at $f(a)$, then the composite function $h = g \circ f$, defined by $h(x) = g(f(x))$, is continuous at a .²

Problem 6. Consider the following function

$$f(x) = \frac{2x^{4/3} - x \sin(x)}{\sqrt{x^2 - 1}}$$

- (a) Where is f defined? Write the answer in terms of intervals.
- (b) Evaluate the derivative f' of f . You should write out all the steps of computing the derivative, and indicate the differentiation rule you are using at each step.

¹Expressions of the form “for all x ” and “there is a y ” are called *quantifiers*. In this exercise we see that these quantifiers cannot be permuted without changing the meaning of a statement.

²Hint: Given $\varepsilon > 0$, take $\delta_1 > 0$ from the continuity of g , then use $\varepsilon_2 = \delta_1$ and the continuity of f to get a δ_2 .